10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 7.4 — Three-Figure Bearings

Unit 7 | Solved Exercise | New Book (PECTAA)

Exercise 7.4 (Solved)

A bearing is always measured clockwise from the North line, written with three figures (015°, 140°, 290°). The bearing of A from B and the bearing of B from A differ by 180°.

1. Find bearing of point P in each of the following:

Bearings from the four diagrams

Diagram	Position of P	Working	Bearing
(i)	40° from due South towards East	180° − 40°	140°
(ii)	20° from due West towards North	270° + 20°	290°
(iii)	50° from due West towards South	270° − 50°	220°
(iv)	75° from due East towards North	90° − 75°	015°

In every diagram the angle is first shifted to the North line, then measured clockwise.

2. The diagram shows the positions of two ships P and Q. (i) What is the bearing of ship P from ship Q? (ii) What is the bearing of ship Q from ship P?

(i) bearing of P from Q

At Q the line QP makes 60° with the North line, on the west side

Bearing of P from Q = 360° − 60° = 300°

Answer: Bearing of P from Q = 300°

(ii) bearing of Q from P

The reverse bearing is 180° different

Bearing of Q from P = 300° − 180° = 120°

Answer: Bearing of Q from P = 120°

3. The diagram shows 3 places A, B and C. Find the bearing of: (i) A from C, (ii) C from A, (iii) C from B, (iv) B from C.

At C the North line divides the angle ACB into 50° (towards A) and 40° (towards B).

(i) A from C

CA is 50° to the west of North, so the bearing is measured the other way

Bearing = 360° − 50° = 310°

Answer: Bearing of A from C = 310°

studysapiens.com

(ii) C from A

Bearing = 310° − 180° = 130°

Answer: Bearing of C from A = 130°

(iii) C from B

CB is 40° to the east of North, so bearing of B from C is 040°

Reverse bearing = 40° + 180° = 220°

Answer: Bearing of C from B = 220°

(iv) B from C

Bearing = 040°

Answer: Bearing of B from C = 040°

4. Abdul Hadi walks 100 m North and then 300 m East. (i) How far is he from his starting position? (ii) On what bearing should he walk to get back to his starting position?

(i) distance from the starting position

The two walks are perpendicular, so use Pythagoras theorem

$d^2 = 100^2 + 300^2 = 10000 + 90000 = 100000$

$d = \sqrt{100000} = 316.2$ m

Answer: Distance from the starting point ≈ 316.2 m

(ii) bearing back to the starting position

$\tan\theta = \frac{100}{300} = 0.3333$

θ = 18.4° (measured from the West line towards the South)

Bearing = 270° − 18.4° = 251.6°

Answer: He should walk on a bearing of about 252°

5. Three ships L, M, N are in the position shown in the diagram. Ship M is North East of ship N. Find the bearing of L from M.

At M the North line makes 60° with the line NM, so the bearing of N from M is 60° + 180° = 240° (which also agrees with M being North East of N).

The angle LMN is marked 90°, and L lies on the left of MN

So L is towards the North-West of M

Bearing of L from M = 240° + 90° = 330°

Answer: Bearing of L from M ≈ 330°

6. A ship sails 20 km on a bearing of 120°. How far to the South has the ship moved from its original position?

Bearing 120° is 30° east of due East, so the South direction makes 60° with the path of the ship.

Distance towards South = 20 cos 60°

= 20 × 0.5000

= 10 km

Answer: The ship has moved 10 km to the South

7. Fatima walked South for 5.5 km and then turned West for 1.3 km. Calculate Fatima's bearing from her starting point.

South distance = 5.5 km, West distance = 1.3 km

$\tan\theta = \frac{1.3}{5.5} = 0.2364$

$\theta = 13.3°$ (measured from due South towards the West)

Bearing = 180° + 13.3° = 193.3°

Answer: Bearing from the starting point ≈ 193°

8. An aircraft flies 150 km east, then 100 km northeast (45° from East). What is the total displacement?

East component = 150 + 100 cos 45° = 150 + 70.71 = 220.71 km

North component = 100 sin 45° = 70.71 km

$\text{Displacement}^2 = 220.71^2 + 70.71^2$

$= 48713 + 5000 = 53713$

Displacement = √53713 = 231.8 km

$\tan\theta = \frac{220.71}{70.71} = 3.121$

$\theta = 72.2°$ (east of North)

Answer: Displacement ≈ 231.8 km on a bearing of about 072°

9. A ship sails 100 km on a bearing of 045°, then changes course and sails 120 km on a bearing of 135°. Find the distance between the starting point and the final position.

First leg: East = 100 sin 45° = 70.71 km, North = 100 cos 45° = 70.71 km

Second leg: East = 120 sin 135° = 84.85 km, North = 120 cos 135° = − 84.85 km

Total East = 70.71 + 84.85 = 155.56 km

Total North = 70.71 − 84.85 = − 14.14 km

$\text{Distance}^2 = 155.56^2 + 14.14^2$

$= 24200 + 200 = 24400$

Distance = √24400 = 156.2 km

Answer: Distance from the starting point ≈ 156.2 km

studysapiens.com